

**B.C.A. (Semester-II) Examination, 2020
(2019-22)**

DESCRETE MATHEMATICS

[Paper Code : BC-201]

Time : Three Hours

[Maximum Marks : 80]

Note: Candidates are required to give their answers in their own words as far as practicable. The questions are of equal value. Answer **any five** questions.

1. Define the following with an example:
 - (a) Null set
 - (b) Universal set
 - (c) Super set
 - (d) Difference of two set
2. List all partition of sets:
 - (a) $A = \{1, 2, 3\}$
 - (b) $B = \{a, b, c, d\}$
3. Given $U(\text{Universal}) = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
 $A = \{2, 4, 6\}$ $B = \{1, 3, 5, 7\}$ $C = \{6, 7\}$. Find:
 - (a) $A' \cap B$
 - (b) $(A \cup B) - C$
 - (c) $(A \cup C)'$
 - (d) $(A \cap U) \cap (B \cup C)$
4. If $f(x) = ax^2 + bx + 2$, $f(1) = 3$, $f(4) = 42$, find b .
5. Find b , if $f(4) = x^4 - 3x^3 + bx^2 + 10x + 16$.
6. If $A = \{1, 3, 5\}$, $XRY \Leftrightarrow Y = X + 2$ and $Y = \{3, 4, 5, 6, 7\}$,
 S be the relation such that $XRY \Leftrightarrow X < Y$.
Then find the following
 - (a) Find $R \circ S$
 - (b) Find $S \circ R$
 - (c) $R \circ S$ and $S \circ R$ via a diagram

7. If $f(z) = \frac{2}{x-2} - \frac{2}{x}$, then find the value of $f(-1.5)$.

8. A tree has $2n$ vertices of degree 1, $3n$ vertices of degree 2 and n vertices of degree 3. Determine the number of vertices and edge in the tree.

9. (a) Use K-map to simplify the following expression:

$$X = A'BC'D' + ABC'D' + A'BCD' + ABCD'$$

- (b) Simplify using K-map.

$$F(A, B, C, D) = \Sigma m(0, 2, 7, 8, 10, 15)$$

10. Define the following terms:

- (a) Power set
- (b) Operations on sets
- (c) Groups
- (d) Rings

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